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Comparison of surface water treatment efficiency using integrated pressure-driven membrane processes and conventional treatment systems

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Abstract: Pressure-driven membrane techniques are increasingly being applied to natural water treatment as an alternative to conventional physico-chemical processes. This makes it essential to compare the effectiveness of these two approaches, particularly under long-term operational conditions. In the preliminary phase of the study, short-term tests were conducted to select the most suitable membranes based on their separation and transport properties: two out of three microfiltration (MF) membranes, two out of three ultrafiltration (UF) membranes, and one out of two nanofiltration (NF) membranes were chosen for subsequent long-term tests. During these long-term evaluations, the efficiency of surface water treatment in integrated MF+NF and UF+NF systems was assessed and compared with the performance of a conventional treatment train comprising coagulation, sedimentation, filtration, and adsorption.

The results indicate that integrated MF+NF and UF+NF systems achieve significantly higher treatment efficiency than conventional processes, with, for example, dissolved organic carbon (DOC) reduction of 93% in membrane systems compared with 50% using conventional methods. Membrane processes also enable highly effective removal of specific microorganisms, which is not achievable using traditional treatment methods. Additionally, integrated MF+NF and UF+NF systems reduce water salinity and achieve water softening, whereas conventional systems tend to increase salinity. An analysis of operational costs revealed that the UF+NF system is nearly 50% less expensive to operate than the MF+NF system. Nevertheless, both membrane-based systems remain several times more costly than conventional water treatment.

Introduction

Freshwater is one of the most important environmental resources, essential for the proper functioning of all living organisms and for all sectors of industry. In Europe, the majority of freshwater is abstracted from surface water bodies: approximately 75% of total water abstraction originates from rivers and reservoirs, while 25% comes from groundwater (Water resources of Europe 2025). Water is primarily used in three main sectors: agriculture, industry, and households. For potable water production, approximately 65% of the supply is derived from groundwater, whereas 35% comes from surface water sources (Europe's water in figures 2027).

Conventional surface water treatment typically involves a sequence of physico-chemical processes designed to remove physical, chemical, and microbiological contaminants. The most commonly applied unit processes include coagulation,

flocculation, sedimentation, rapid sand filtration, adsorption on activated carbon, and disinfection. These conventional treatment methods have been successfully used for decades, particularly in municipal water supply systems. However, they present several limitations in the context of contemporary environmental, health, and technological challenges. Their effectiveness depends strongly on the characteristics and concentrations of pollutants, as well as on the composition of the water matrix (Knap-Baldyga, and Zubrowska-Sudoł 2023). Moreover, these processes generate sludge and backwash water that require appropriate management and disposal (Zielina and Dąbrowski 2021). Coagulation involves the addition of chemical reagents, the residuals of which may contribute to secondary pollution (García-Avila et al. 2025). Similarly, chemical disinfection may lead to the formation of disinfection by-products (DBPs), some of which are potentially toxic or carcinogenic (Srivastav et al. 2020).

In response to the limitations of conventional technologies, low-pressure membrane processes began to be widely implemented in the 1990s (Li & Visvanathan 2017). Compared with conventional surface water treatment, microfiltration (MF) and ultrafiltration (UF) offer several advantages, including stable production of high-quality treated water, reduced or no chemical consumption, compact system design, and ease of automation (Aziz et al. 2024). These processes are particularly effective in removing protozoa (e.g., *Cryptosporidium*), whereas their efficiency in removing bacteria and, especially, viruses may be limited (Bodzek et al. 2019). Furthermore, low-pressure membrane processes are generally ineffective in removing low-molecular-weight organic compounds of both natural and anthropogenic origin, which may contribute, among other effects, to water color.

Nanofiltration (NF) has also been applied in drinking water treatment. Due to the specific separation properties of NF membranes, this process enables the removal of low-molecular-weight organic compounds, including emerging contaminants, and allows partial water softening and salinity reduction (Guo et al. 2022). However, membrane fouling remains a major operational challenge in NF system design and operation, affecting both the separation and transport properties of the membranes. Counteracting this phenomenon or mitigating its intensity significantly impacts the operational costs of membrane processes. As stated by Hong and Elimelech (1997), the presence of natural organic matter (NOM) in water causes calcium ions, responsible, among other things, for water hardness, to form a dense gel layer with high resistance on the NF membrane surface, thereby intensifying membrane fouling. To mitigate fouling intensity in NF systems, primarily through the removal of NOM, pretreatment using either conventional processes (Kim et al. 2011) or low-pressure membrane techniques is widely applied (Liu et al. 2021).

The application of membrane separation processes in drinking water treatment is limited by operational costs,

particularly those associated with energy consumption and membrane fouling. Energy consumption in membrane separation processes has been reduced through the use of increasingly advanced (modified) membranes. The average energy consumption of the UF process is 0.29 kWh/m³ (Lidén and Persson 2016). In the case of NF, it is approximately 1 kWh/m³ (Fan et al. 2020), which is nearly 50% lower than that of RO. For this reason, among others, NF is widely used for freshwater treatment when only softening is required. For comparison, the energy consumption of conventional water treatment systems ranges from 0.6 to 0.8 kWh/m³, depending on the plant size and the type of filtration used (Skoczko 2025).

Regardless of whether conventional physico-chemical processes or membrane-based technologies are employed for water purification, treated surface water must undergo chemical disinfection to prevent secondary microbial contamination within the distribution system. The effectiveness of disinfection and, in particular, the formation and speciation of DBPs are strongly influenced by the concentration and characteristics of organic matter remaining in the treated water (Li et al. 2017; Pietrzyk and Papciak 2017). Numerous studies have shown that the amount of disinfection by-products formed is proportional to the UV₂₅₄ absorbance value (Valenti-Quiroga et al. 2024; Wang et al. 2024). Therefore, the effective removal of organic matter in processes preceding disinfection is crucial to ensuring water safety.

The presence and type of organic matter also affect the degree of membrane fouling and, consequently, the reduction of permeate flux. In particular, the structure of NOM has been indicated as the main factor influencing the extent of membrane fouling (Zularisam et al. 2006). Additionally, membrane durability and permeate quality are determined by operational conditions, including the method and frequency of backwashing and/or chemical cleaning procedures.

The analyses of the effectiveness of membrane separation processes and conventional treatment systems presented in the

Table 1: Raw water characteristics

Parameter	Unit	Min value	Max value
Temperature	°C	12.0	17.4
Conductivity	µS/cm	563	740
Suspended solids	mg/L	1.6	16.6
Turbidity	NTU	2.9	17.8
Color	mg Pt/L	8.96	13.66
UV ₂₅₄	m ⁻¹	9.4	16.87
TOC	mg C/L	4.05	4.99
DOC	mg C/L	3.96	4.92
BDOC	mgC/L	0.31	1.39
Total hardness	mg CaCO ₃ /L	161	290
SO ₄ ²⁻	mg/L	27.1	86.4
Cl ⁻	mg/L	34.1	50.8
<i>Clostridium perfringens</i>	CFU/100 cm ³	0	78
<i>E. coli</i>	CFU/100 cm ³	0	25
Coliform bacteria	CFU/100 cm ³	3	2420
Total cell count (TCC)	cell/mL	37,971	293,069

literature do not allow for a direct comparison of these two approaches, as the studies were conducted using different water sources. In contrast, the present study was carried out in parallel using two treatment systems processing the same water, making it possible to compare both treatment efficiency and operational costs.

Therefore, the aim of this study was to compare the efficiency and costs of surface water treatment using conventional physico-chemical processes with those of integrated systems combining low-pressure membrane processes (microfiltration or ultrafiltration) and nanofiltration. The research was conducted under long-term operational conditions and included membranes made of different materials, enabling the identification of those with the most favorable separation and transport properties.

Materials and Methods

Raw water

The study was conducted to compare the effectiveness of surface water treatment using pressure-driven membrane processes (microfiltration, ultrafiltration, and nanofiltration) with a conventional treatment system comprising sequential coagulation, sedimentation, rapid sand filtration, and granular activated carbon adsorption. The water used in the study was sourced from a surface water body supplying a town in south-western Poland. The feedwater for each treatment system is characterized in Table 1.

Membrane processes

The membrane process tests were conducted in two stages. The first stage aimed to evaluate the separation and transport properties of microfiltration, ultrafiltration, and nanofiltration membranes. Preliminary tests were performed

using three types of microfiltration modules, three types of ultrafiltration modules, and two types of nanofiltration modules. In the second stage, the membranes selected during the preliminary tests in each category were employed in long-term experiments to assess the suitability of integrated MF+NF and UF+NF processes for water treatment. The application of integrated MF+NF and UF+NF systems resulted from the need for partial softening of treated freshwater without complete demineralization.

All membrane modules used in the study were supplied by PolymemTech (Warsaw, Poland). The characteristics of the membrane modules are summarized in Table 2.

Preliminary membrane tests were conducted over a period of 5 hours, while the long-term experiments using sequential systems were carried out for 71 days (MF+NF) and 90 days (UF+NF). All short-term experiments were performed in duplicate. Prior to the main experiments, the membranes were conditioned at the operating pressure for 45 minutes using tap water. During filtration cycles with capillary and tubular modules, the membranes were backwashed every 10 minutes for 1 minute. At the end of each filtration cycle, the membranes were chemically cleaned with a 1% citric acid solution.

Filtration with MF and UF membranes was preceded by pre-treatment of the raw water using 20 µm bag filters. For the NF tests, the feedwater was pre-treated using the MF3 microfiltration membrane.

In the long-term experiments, to minimize membrane fouling, the MF and UF modules were supplied with feedwater pre-treated using a 20 µm bag filter, a UV lamp, and a 10 µm cartridge filter. The ceramic micro- and ultrafiltration membranes were backwashed with permeate every 10 minutes for 1 minute and chemically cleaned weekly with NaOCl (200 mg Cl₂/L) and citric acid (pH 2.8).

Table 2: Membrane modules characteristics

Process	Microfiltration		
module symbol	MF1	MF2	MF3
membrane material	polypropylene (PP)	polypropylene (PP)	ceramic (α Al ₂ O ₃)
mean pore diameter, µm	0.2	0.2	0.2
module configuration	spiral wound	capillary	tubular
membrane surface, m ²	1.8	0.5	1.26
Process	Ultrafiltration		
module symbol	UF1	UF2	UF3
membrane material	modified polytetrafluoroethylene (TFM)	polyethersulfone (PES)	ceramic
cut-off, kDa	1.5-3.5	20	50
module configuration	spiral wound	spiral wound	tubular
membrane surface, m ²	2.4	1.8	0.7
Process	Nanofiltration		
module symbol	NF1	NF2	
membrane material	polyamide (PA)	modified polytetrafluoroethylene (TFM)	
cut-off, Da	200	200-300	
module configuration	spiral wound	spiral wound	
membrane surface, m ²	7.9	7.9	

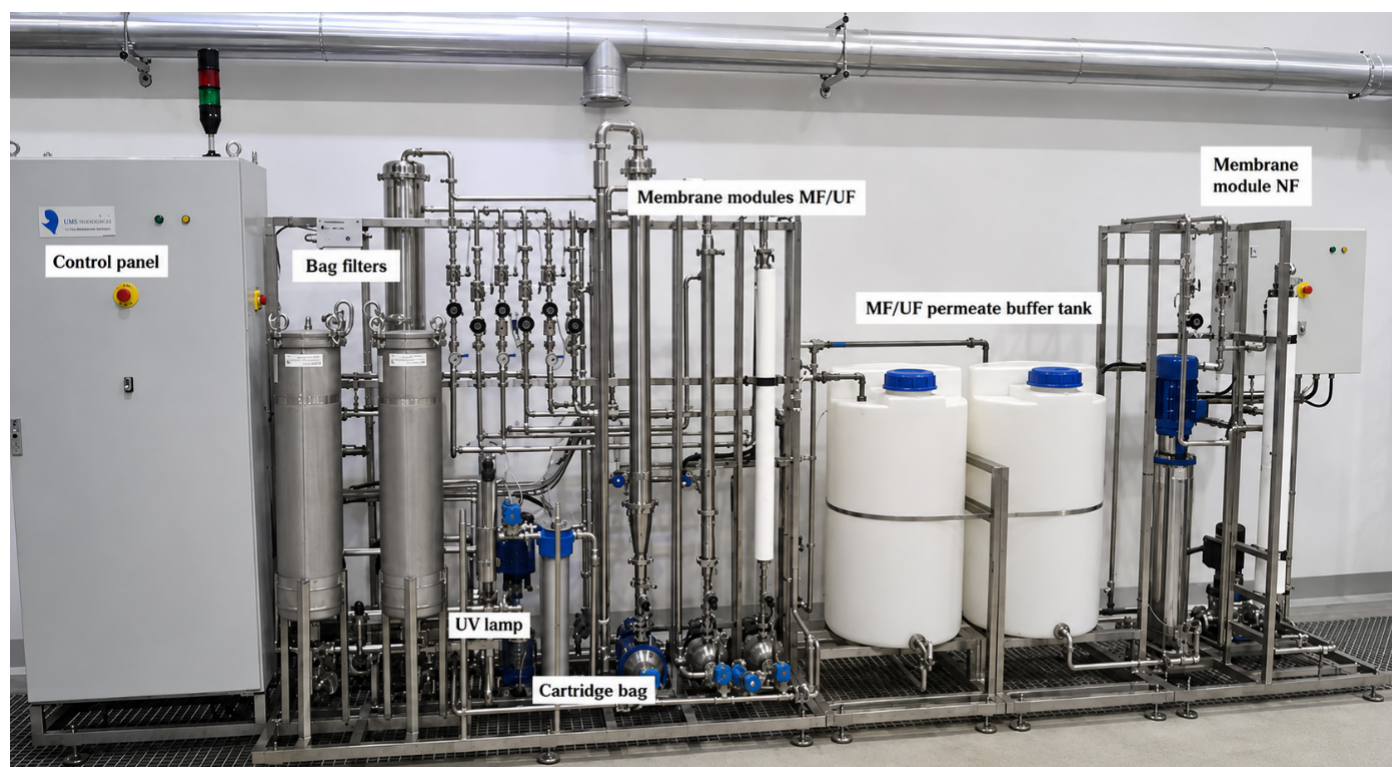


Fig. 1. Experimental set-up of the membrane system

All membrane processes were operated at constant pressures as follows:

- Short-term tests: 1.3 bar for MF, 3.05 bar for UF, and 3 bar for NF;
- Long-term tests: 1.9 bar for MF, 3 bar for UF, and 3.2 bar for NF.

The parameters applied in the short-term and long-term tests were determined based on preliminary studies and the operating conditions achievable within the pilot-scale installation.

All experiments were performed using the setup shown in Figure 1.

Conventional water treatment train

An experiment to evaluate the effectiveness of surface water treatment using a conventional treatment system was conducted in a continuous-flow system with a capacity of 2.95–3.05 m³/h, as illustrated in Figure 2. The treatment train

consisted of coagulation, sedimentation, rapid sand filtration, and adsorption on granular activated carbon (GAC), with the operating parameters summarized in Table 3.

The system was operated continuously, and the process parameters remained largely stable throughout the study, except for minor adjustments in reagent dosages. The activated carbon was loaded into the filters two months prior to the start of the experiments, allowing it to develop biological activity.

Analytical methods

To evaluate the effectiveness of the individual treatment processes, the following parameters were measured in the raw water and in samples collected after each process step: temperature, conductivity, color, turbidity, suspended solids concentration, absorbance at 254 nm, dissolved organic carbon (DOC) concentration, total hardness, sulphate and chloride ion concentrations, and the presence of coliform bacteria, *Escherichia Coli* (*E. coli*), and *Clostridium perfringens*. All

Table 3: Operational parameter of conventional unit processes

Process:	Contact time (min)	Reagent	Dose, mg/L
Coagulation:		polyaluminium chloride of 65% basicity	1.94-2.89
Rapid mixing	2-3		
Flocculation	17-21		
Sedimentation	115-125	-	-
Sand filtration	18-25 (rate 5 m/h)	sand	-
Adsorption by GAC	18-25 (rate 5 m/h)	GAC WG-12 (specific surface area 968 m ² /g)	-
Stabilisation	-	H ₂ SO ₄	2.1-3.5
Disinfection	-	NaOCl	0.7-1.0

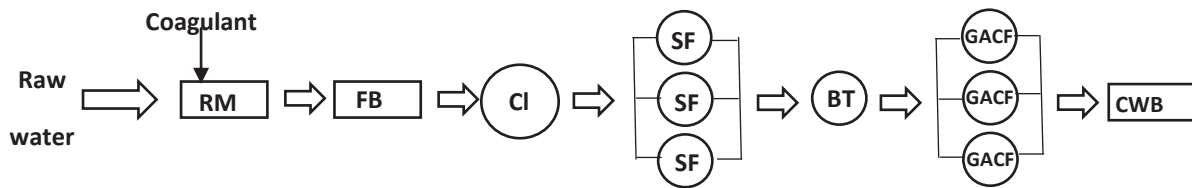


Fig. 2. Experimental set-up of the conventional water treatment train (RMT – rapid mixing tank, FB – flocculation basin, CI – clarifier, SF – sand filter, BT- buffer tank, GACF – granular activated carbon filter, CWB – clean water basin)

analyses were conducted in accordance with current standard methods. Based on UV_{254} and DOC measurements, the specific UV absorbance (SUVA) was calculated. The analytical error of the performed measurements did not exceed 2%.

The molecular weight distribution of organic substances was determined using size-exclusion chromatography (SEC), following the procedure described in detail by Allpike et al. (2005). The SEC results were also presented as three-dimensional UV absorbance spectra (3D spectra) over the full wavelength range, allowing inferences regarding the properties of the organic substances.

For the long-term evaluation of both the membrane processes and the conventional treatment system, cytometric analyses were performed to determine total cell counts (TCC), the proportion of damaged cells, and the concentration of biodegradable dissolved organic carbon (BDOC). DNA staining was conducted using SYBR Green (SG) to quantify TCC and propidium iodide (PI) to assess damaged cells. Cytometric measurements were carried out using a BD Accuri™ C6 flow cytometer (BD Biosciences, USA).

Operational costs analysis

The comparison of the conventional water treatment system with the integrated membrane systems also included an analysis of the operational costs of each technological configuration.

Operational costs were determined based on measurements conducted during the study. The systems were equipped

to monitor energy consumption, the volume of water used for rinsing, and the amount of reagents used for membrane cleaning. These measurements were used to determine the average consumption of energy, reagents, and consumable materials. Costs were then calculated using the unit prices of the individual materials applicable during the study period and incurred by the company. Unit prices were based on the company's supply contracts for reagents, energy, and other consumable materials. For the integrated membrane processes, the operational costs comprised:

1. Pre-treatment, including the operating costs of the bag filter system, cartridge filters, and the water heater and/or UV disinfection lamp.
 2. Operation of MF, UF, and NF membranes, covering energy consumption and the costs associated with membrane rinsing (periodic chemical cleaning costs were not included).
 3. Idle operation, encompassing the control and electrical systems of the installation.
 4. Membrane replacement, assuming a service life of five years.
- For the conventional treatment system, operational costs included:

1. Energy consumption for water pumping and mixing, covering the power supply for pumps, mixers, and reagent dosing pumps.
2. Reagent consumption, including coagulants, sodium hydroxide for pH adjustment, sodium hypochlorite, oxygen for ozone generation, and activated carbon.

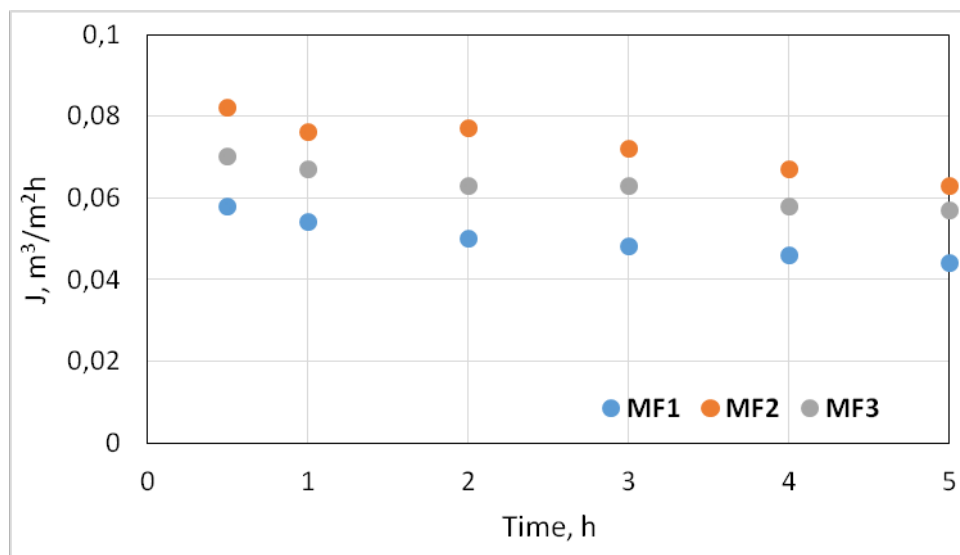


Fig. 3. Short-term permeate flux profiles for microfiltration membranes

Average reagent dosages used during system operation were adopted for cost estimation.

3. Filter backwashing, including the energy costs of blowers and pumps used for water and air backwashing.
- All costs were estimated based on 2025 prices.

Results and discussion

Evaluation of the transport and separation properties of micro-, ultra-, and nanofiltration membranes: a short-term study

To preliminarily assess the suitability of the membranes, their transport and separation properties were evaluated in short-term experiments. Analysis of the transport performance of the microfiltration modules showed that permeate flux was influenced by both the membrane material and the module configuration and operating mode (Figure 3, Table 4).

In the case of microfiltration membranes, which, according to the literature (Mulder 1996), are the most susceptible to fouling because they exhibit the highest convective transport among the three membrane types investigated (MF, UF, NF), detailed measurements of permeate flux were performed to analyze the kinetics of this phenomenon. Analysis of the permeate flux decline profiles over time showed that none of the membranes exhibited a rapid initial drop in hydraulic performance, which would indicate complete pore blocking. Instead, all membranes exhibited a gradual decline in performance throughout the experiment, indicating a mixed-mode fouling mechanism (intermediate blocking, adsorption, and cake layer formation).

For the polypropylene (PP) membranes, a significantly higher permeate flux was recorded in the capillary module (MF2) than in the spiral-wound module (MF1). This difference resulted from the fact that the capillary module was operated with periodic backwashing, whereas backwashing was not possible in the spiral-wound configuration. Despite the lower absolute permeate flux observed for the membrane in module MF1 compared with MF2, the relative decline in hydraulic performance after 5 hours of filtration was comparable for both systems, amounting to 23.8% and 27.9% for MF1 and MF2, respectively.

A comparison of the transport properties of PP and ceramic membranes, both operated with systematic backwashing (modules MF2 and MF3), revealed higher hydraulic performance for the polymeric membrane. This finding is somewhat unexpected, as PP membranes are generally

considered hydrophobic (Ariono and Kusuma Wardani 2017), whereas ceramic membranes are typically regarded as strongly hydrophilic (Gu et al. 2022). Polypropylene is widely used as a membrane material due to its chemical stability, low cost, and good mechanical properties. Moreover, surface modification can render PP highly hydrophilic (Gryta 2021), which may lead to an increased permeate flux during the filtration of aqueous solutions. Although no information was available from the manufacturer of the membrane used in module MF2 regarding its surface treatment, it can reasonably be assumed that the membrane had undergone hydrophilization.

Analysis of the transport properties of the ultrafiltration membranes indicated that the highest permeate flux was achieved with the ceramic membrane installed in the tubular UF3 module, whereas the lowest flux was recorded for the TFM membrane in the spiral-wound UF1 module. The observed differences are attributable to both the membrane material (PTFE, PES, and ceramic) and the substantially different molecular weight cut-off (MWCO) values: 1.5–3.5 kDa for UF1, 20 kDa for UF2, and 50 kDa for UF3. Moreover, the tubular membrane in the UF3 module was subjected to systematic backwashing, which effectively limited the formation of a fouling layer and contributed to maintaining a higher permeate flux.

It is also noteworthy that, despite the absence of backwashing during the operation of the UF1 module (MWCO 1.5–3.5 kDa), the permeate flux remained nearly constant throughout the 5-hour test. This suggests that the retained particles were sufficiently large to prevent internal pore blocking and were continuously removed from the membrane surface by the circulating feed stream, thereby minimizing fouling.

For the nanofiltration membranes, both systems exhibited stable permeate flux throughout the experimental period. However, superior transport performance was observed for the NF1 membrane. Although the two tested membranes have similar molecular weight cut-off values, they differ in terms of membrane material. The membrane in the NF1 module was fabricated from polyamide, a highly hydrophilic polymer (El-Gendi 2022), which may account for its higher hydraulic performance.

Furthermore, no decline in permeate flux was recorded for either membrane during the short-term tests, indicating the absence of fouling under the applied operating conditions. This favorable performance may also be attributed to the pre-treatment of the feed solution using a ceramic microfiltration

Table 4: Permeate flux for membranes tested in short-term experiments

	Permeate flux, m ³ /m ² h			
	min	max	mean	SD
MF1	0.043	0.056	0.048	0.0039
MF2	0.067	0.093	0.078	0.0066
MF3	0.054	0.071	0.063	0.005
UF1	0.028	0.028	0.028	0
UF2	0.049	0.069	0.058	0.0079
UF3	0.136	0.152	0.143	0.0075
NF1*	0.013	0.013	0.013	0
NF2*	0.009	0.009	0.009	0

* preceded by MF3

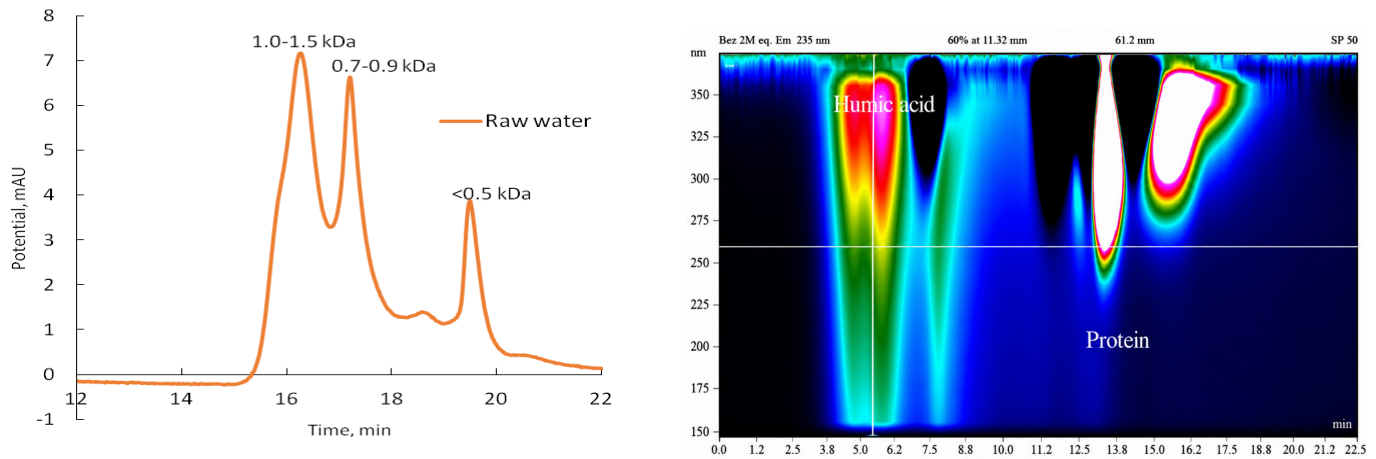


Fig. 4. Particle size distribution (a) and 3-D absorbance spectra (b) for the surface water

membrane, which likely reduced the fouling potential of the feed stream.

The selection of membranes for further long-term testing was based not only on their transport properties but, above all, on their effectiveness in removing specific groups of pollutants from the surface water.

The raw water was characterized by a moderate concentration of organic matter ($\text{DOC} < 5 \text{ mg C/L}$). Size exclusion chromatography (SEC) analysis indicated that the organic fraction consisted primarily of humic acids with molecular weights of approximately 1 kDa and proteins with molecular weights below 0.15 kDa (Figure 4). These findings are consistent with the SUVA values, which ranged from 2.4 to 3.4 $\text{L/mg} \cdot \text{m}$, suggesting the presence of a mixture of humic substances with varying molecular weights.

As shown in Table 5, among the microfiltration membranes tested, the best water treatment performance was achieved with the MF3 ceramic membrane. All membranes provided more than 90% removal of suspended solids and more than 95% turbidity reduction. Since particles larger than 5 μm are primarily responsible for water turbidity (Folorunso et al. 2022), such high turbidity removal efficiencies were expected for all tested membranes. Moreover, the removal of suspended solids and turbidity remained stable throughout each filtration cycle.

In contrast, the removal efficiency of organic matter by the microfiltration membranes was significantly lower. It did not exceed 35% for color, 19% for UV absorbance at 254 nm, 18% for TOC, or 17% for DOC. Similarly low organic matter removal efficiencies for microfiltration have also been reported by Metsämuuronen et al. (2014). This limited performance results from the relatively large membrane pore size (0.2

μm) and the low molecular weight of the organic compounds present in the treated water.

At the same time, for the parameters characterizing organic matter content (TOC, DOC, color, and UV_{254}), a gradual increase in separation efficiency was observed over the course of the process. This effect was most likely associated with membrane fouling and, in particular, with the formation of an additional dynamic (cake) layer on the membrane surface. A similar phenomenon was reported by Celik et al. (2022).

Ultrafiltration tests (Table 5) demonstrated that only the dense UF1 membrane enabled significant removal of organic matter, achieving approximately 76% reduction in DOC, 82% reduction in color, and 75% reduction in UV_{254} absorbance. Particle size distribution analysis of the raw water (Fig. 4a) indicated that compounds with molecular weights in the range of 0.9–1.5 kDa predominated. Considering the manufacturer-declared molecular weight cut-off (MWCO) of 1.5–3.5 kDa for the UF1 membrane, these compounds were removed with high efficiency.

Spectrophotometric analysis (Fig. 4b) further confirmed the presence of humic acids in the raw water, in addition to proteins and fulvic acids. Humic acids were effectively retained by the dense UF membrane. The observed decrease in color and UV_{254} absorbance is directly related to the removal of humic substances, which contain aromatic rings and conjugated double bonds in their structure (Gomes de Melo et al. 2016).

In contrast, the UF2 and UF3 membranes achieved more than 90% removal of suspended solids and turbidity; however, their efficiency in removing organic matter was comparable to that observed for the microfiltration membranes. For example, TOC and DOC removal did not exceed 20%. The MWCO

Table 5: Removal efficiency of selected water quality parameters using microfiltration and ultrafiltration membranes

Parameter	Retention coefficient, %					
	MF1	MF2	MF3	UF1	UF2	UF3
Turbidity	95.9-97.4	97.5-98.1	98.4-99.2	89.6-97.1	94.4-97.6	89.6-97.1
TOC	6.3-7.7	8.6-13.6	11.2-17.7	74.1-78.0	9.2-22.5	14.9-19.4
DOC	5.1-7.0	7.8-10.7	10.5-17.0	74.1-77.6	14.1-20.4	13.4-18.6
Color	26.4-34.8	27.5-33.2	25.7-34.1	79.6-86.1	26.8-33.5	20.6-26.8
UV_{254}	11.2-13.4	9.7-12.5	12.1-19.1	70.7-76.5	13.1-18.8	10.7-15.1

Table 6: Removal efficiency of selected water quality parameters using nanofiltration membranes

Parameter	Retention coefficient, %	
	NF1	NF2
TOC	93.2-96.0	96.1-97.4
DOC	93.3-95.9	95.9-97.4
Color	94.9-96.8	96.1-96.6
UV ₂₅₄	88.0-96.5	94.9-96.1
Hardnes	29.4-33.3	31.7-33.0
Conductivity	39.9-48.2	39.3-47.0
Cl ⁻	3.9-9.7	6.4-11.2

values of these membranes were 20 kDa (UF2) and 50 kDa (UF3), which, given the relatively low molecular weight of the organic compounds present in the treated water, explains their limited separation performance. Similar conclusions were reported by Prihartini Aryanti et al. (2025), who demonstrated that only tight ultrafiltration membranes are effective for the removal of natural organic matter (NOM) from surface waters.

During the short-term experiments with nanofiltration modules, the surface water was pretreated using the MF3 ceramic microfiltration membrane. As a result, the feed stream supplied to the NF modules was practically free of suspended solids.

As shown in Table 6, both nanofiltration membranes achieved organic matter removal efficiencies exceeding 90%, with slightly higher retention observed for the NF2 membrane. In addition, nanofiltration reduced water hardness by more than 30%. SEC analysis of the NF permeate (Figure 5) indicated that it mainly contained proteins with molecular weights below 0.15 kDa.

The high efficiency of organic matter removal is consistent with findings of other researchers (Liu et al. 2021, Yuksekdag et al. 2023), who have demonstrated that organic macromolecules, due to their relatively high molecular weight (compared to the MWCO of NF membranes) and their negative charge, are effectively retained during nanofiltration.

Although the NF1 and NF2 membranes have similar MWCO values, they differ in membrane material: NF1 is made of polyamide, whereas NF2 is fabricated from modified polytetrafluoroethylene (PTFE). The latter material is more hydrophobic, which may contribute to a lower permeate

flux but, at the same time, may reduce convective transport of organic compounds through the membrane, potentially enhancing their retention.

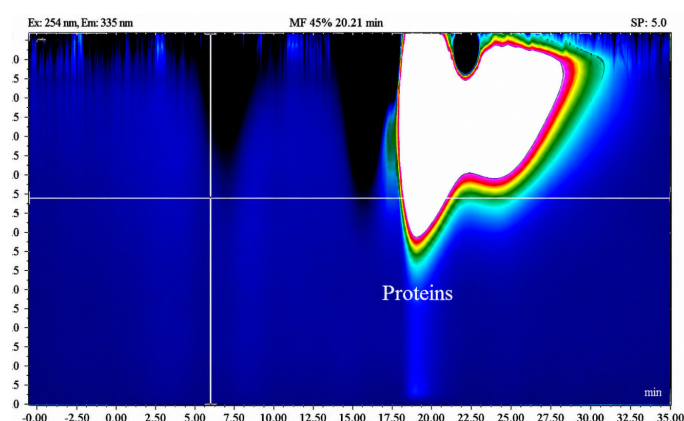
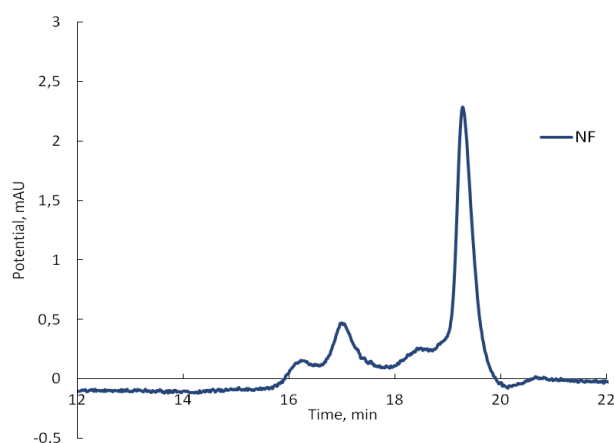
Transport and separation properties of micro-, ultra-, and nanofiltration membranes in long-term testing

Given that the primary objective in selecting the treatment technology was to produce high-quality treated water, membranes for long-term experiments were selected primarily on the basis of their separation performance. The results of the short-term experiments indicated that the MF3 microfiltration, UF3 ultrafiltration, and NF2 nanofiltration membranes exhibited the best separation properties, and these were therefore selected for extended testing.

Before assessing the overall surface water treatment efficiency of the integrated membrane systems, long-term experiments were conducted using both the best-performing membranes (MF3 and UF3) and membranes with lower separation efficiency (MF1 and UF2). As shown in Figure 6, the separation performance of both micro- and ultrafiltration membranes remained largely unchanged throughout the test period. This stability was observed for membranes subjected to backwashing (MF3 and UF3) as well as for those operated without this cleaning procedure (MF1 and UF2).

For turbidity removal, the microfiltration membranes achieved average reductions of 92.5% (MF1) and 90.5% (MF3), while the ultrafiltration membranes reduced turbidity by 97.1% (UF2) and 95.4% (UF3). In contrast, the low molecular weight of the organic compounds present in the water resulted in limited removal during long-term experiments: DOC reduction reached only 17.2%, 27.0%, 31.1%, and 29.7% for MF1, MF3, UF2, and UF3, respectively. Among the removed organic matter, nearly half was biodegradable, with average BDOC removal efficiencies of 44.7%, 43.6%, 46.3%, and 44.8% for MF1, MF3, UF2, and UF3, respectively.

Throughout the long-term experiments, all tested low-pressure membranes achieved complete (100%) retention of *Escherichia coli* and *Clostridium perfringens* spores. Similarly, the removal efficiency of total cell counts (TCC) was close to 100%, regardless of membrane type. Nevertheless, some bacteria were still detected in the permeate, with the microfiltration membranes allowing several times more bacterial passage than the ultrafiltration membranes

**Fig. 5.** Particle size distribution (a) and three-dimensional absorbance spectra (b) of NF2 nanofiltration permeate

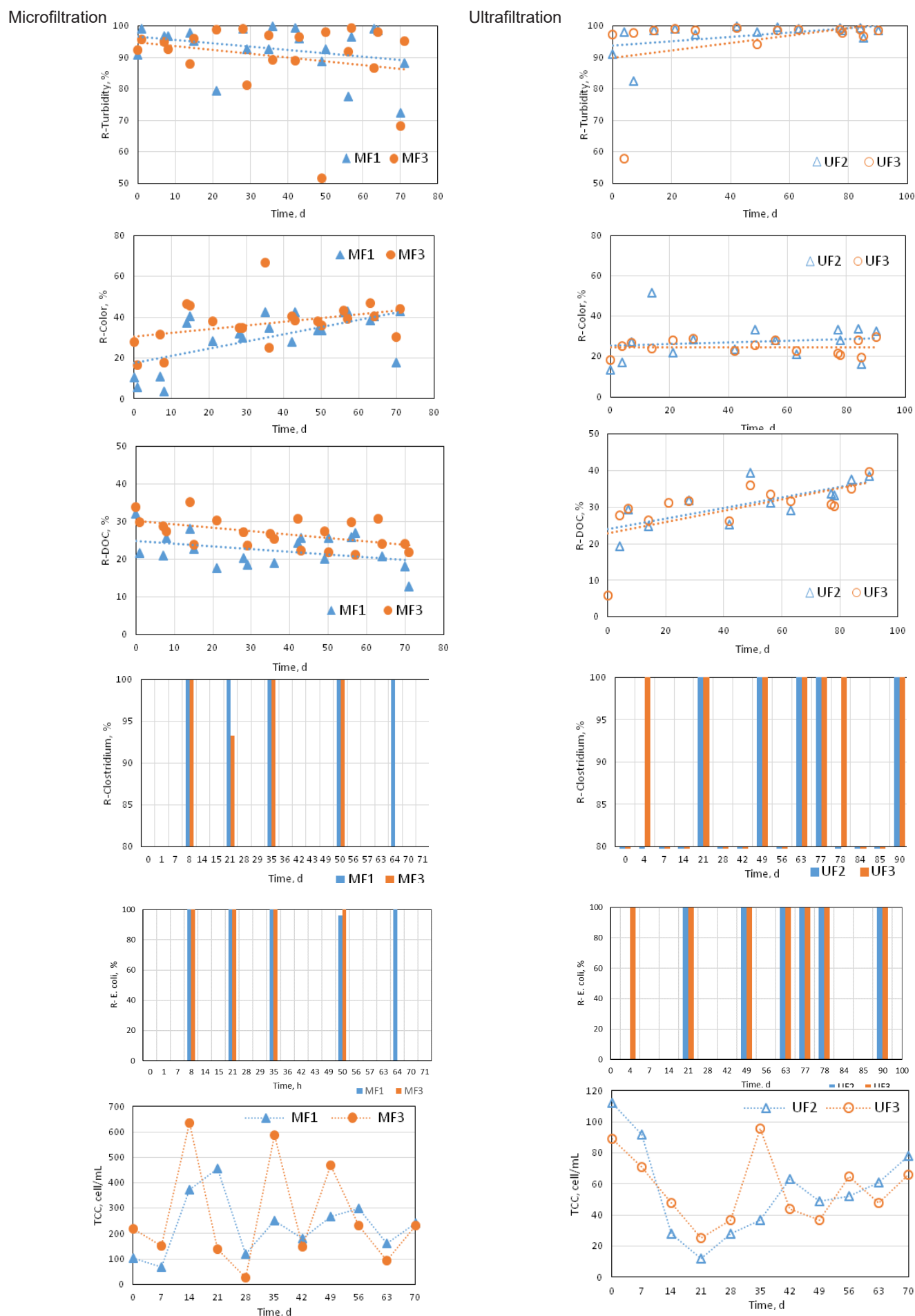


Fig. 6. Separation properties of micro- and ultrafiltration membranes during long-term experiments

(Figure 6). The significantly higher bacterial removal efficiency observed for both ultrafiltration membranes may help reduce the required disinfectant dosage in subsequent stages of water treatment.

Long-term testing with the MF and UF membranes confirmed that MF3 and UF3 exhibit the best separation performance. Consequently, in evaluating the effectiveness of surface water treatment in an integrated system, these membranes were used for pretreatment prior to filtration with the NF2 nanofiltration membrane. Regardless of the pretreatment applied, the nanofiltration process ensured water softening, which was the primary objective of incorporating nanofiltration into the treatment train. At the same time, the reduction in salinity (degree of demineralization) resulting from NF did not necessitate remineralization of the treated water, which may be required when reverse osmosis (RO) is used. (Gao et. al. 2022).

These experiments also enabled a comparison of the performance of the integrated membrane system and that of a conventional treatment system comprising coagulation, sedimentation, rapid sand filtration, and activated carbon adsorption. Long-term experiments on surface water treatment using integrated MF3–NF2 and UF3–NF2 processes (Figure 7) showed that, during the initial stage of operation, pretreatment with the UF membrane provided higher hydraulic performance of the NF2 membrane, although this advantage diminished over time. In contrast, when nanofiltration was preceded by microfiltration, the permeate flux of the NF2 membrane remained nearly constant throughout the experiment.

A comparison of the surface water treatment efficiency of the integrated membrane processes with that of the conventional treatment system (Figure 8) shows that, over long-term experiments, the membrane processes achieved substantially higher removal of organic matter. Specifically, average reductions of more than 95% in color, 93% in DOC, and 93% in UV_{254} absorbance were observed, compared with 76%, 50%, and 61%, respectively, for the conventional treatment system.

Long-term experiments also demonstrated that the

treated water quality in the MF+NF and UF+NF systems was comparable, regardless of whether microfiltration or ultrafiltration was used as the low-pressure membrane pretreatment step. In the conventional treatment system, despite confirmed microbiological activity in the adsorption beds, the removal of biodegradable dissolved organic carbon (BDOC) was nearly 50% lower than that achieved by membrane systems, with similar results obtained regardless of the type of low-pressure membrane used for pretreatment.

Significant differences between the treatment systems were also observed in terms of total cell count, which directly affects the required disinfectant dosage. Only the integrated UF+NF system reduced cell counts to a level sufficient to ensure biological stability. In contrast, water treated by the conventional system contained 15 and 5 times more cells than water treated by the UF+NF and MF+NF systems, respectively.

Furthermore, treatment using integrated membrane systems achieved average reductions of 21% in total hardness and 76% in sulfate concentration. The removal of these inorganic components by nanofiltration also led to a substantial decrease in water conductivity—43% for MF+NF and 45% for UF+NF, which could not be achieved with the conventional treatment system. As shown in Figure 9, none of the unit processes in the conventional system effectively removes inorganic ions, and coagulation could even increase water salinity. For example, the use of polyaluminium chloride as a coagulant led to a 6% increase in conductivity.

The study also enabled a comparison of the removal efficiency of different fractions of organic compounds present in the water. Particle size distribution analysis (Figure 10) showed that, following conventional water treatment, the relative proportions of the individual fractions remained similar to those in the raw water. Treatment with low-pressure membrane processes slightly reduced the overall organic matter content and partially removed the larger fractions with molecular weights above 1.0–1.5 kDa. In contrast, nanofiltration effectively retained most of the organic matter, allowing only low-molecular-weight natural organic matter (NOM) with molecular weights below 0.15 kDa to remain in

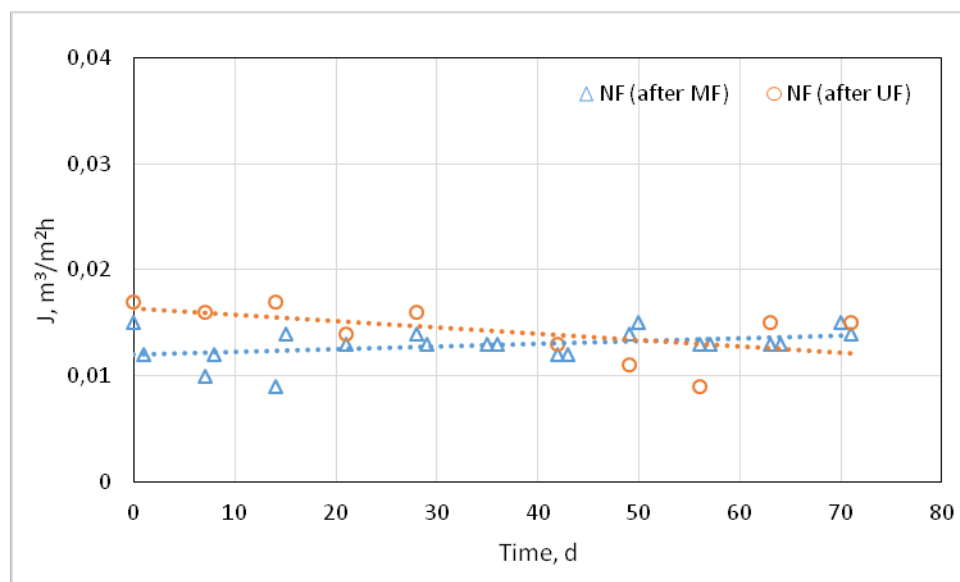


Fig. 7. Effect of surface water pretreatment on NF membrane permeate flux during long-term tests

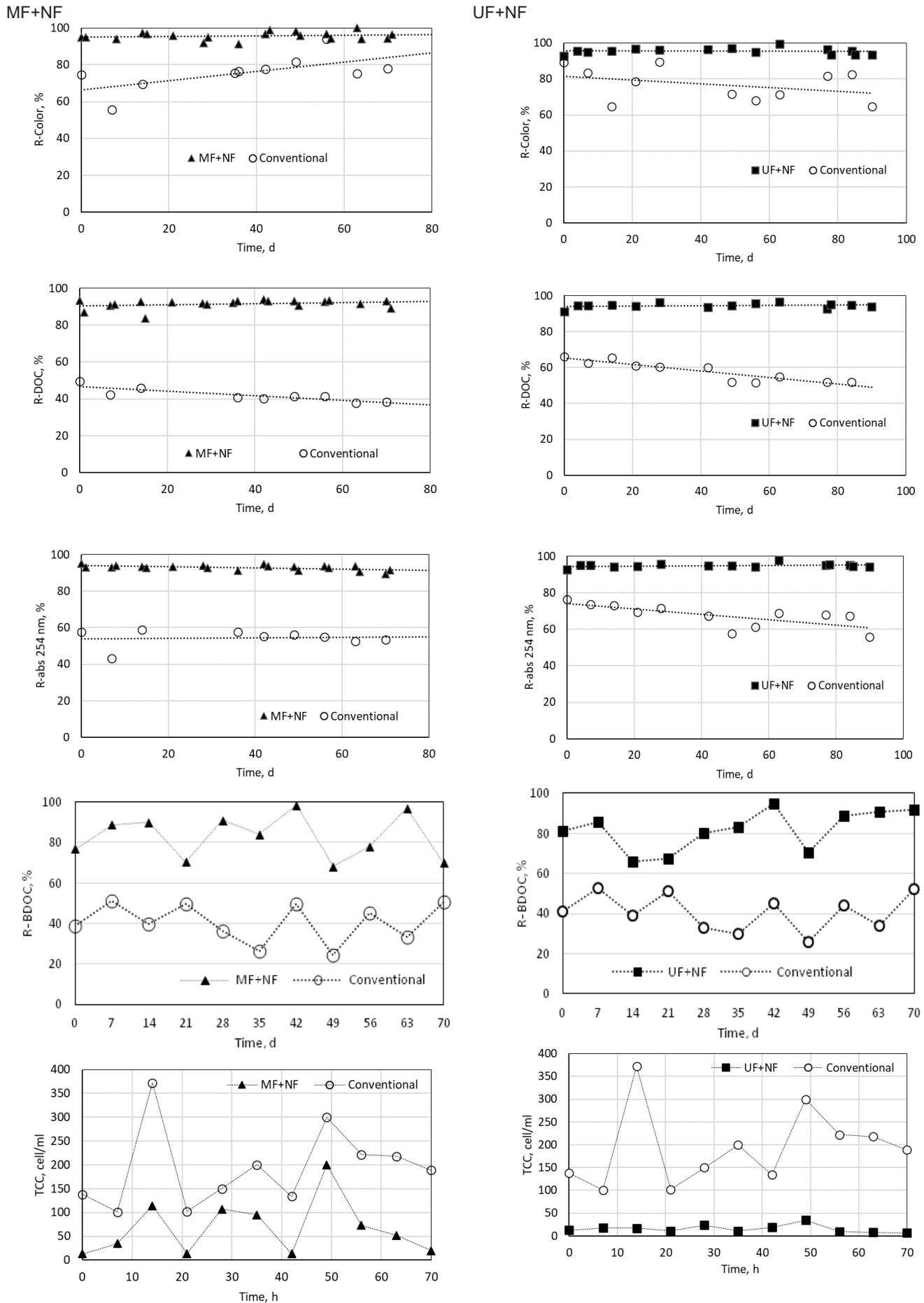


Fig. 8. Comparison of water treatment efficiencies of MF+NF and UF+NF membrane processes with a conventional system comprising coagulation, sedimentation, rapid sand filtration, and granular activated carbon adsorption

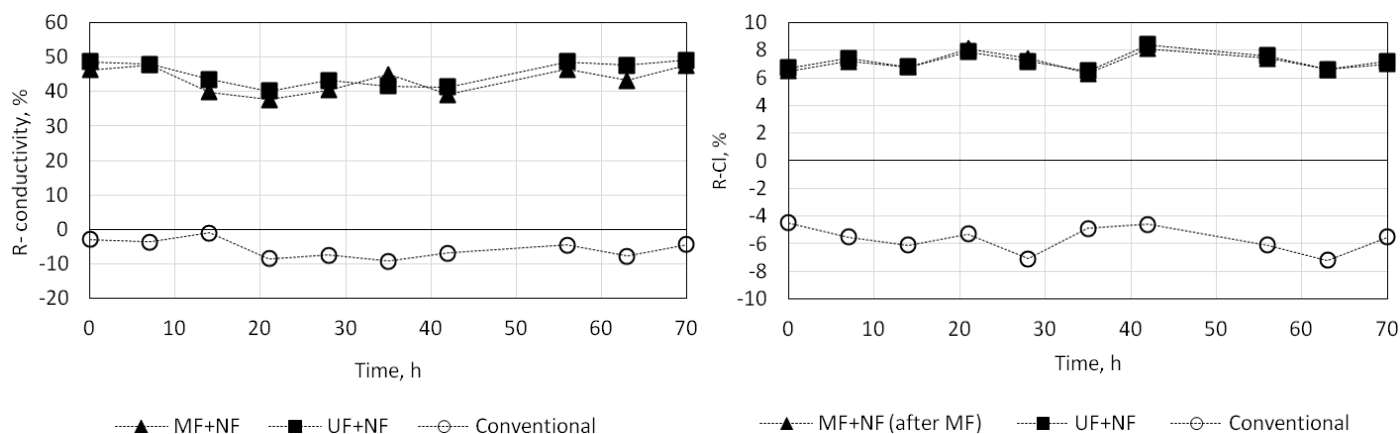


Fig. 9. Effect of water treatment method on conductivity (a) and chloride ion concentration (b)

the permeate.

The substantial effect of nanofiltration on the removal of medium- and high-molecular-weight organic substances is reflected in the average concentrations of the individual fractions in the feed water and in the treated water from each treatment system. As shown in Figure 11, pretreatment with ultrafiltration prior to nanofiltration enhanced the removal of particles with molecular weights above 0.7–0.9 kDa. In contrast, the conventional treatment system achieved average removal efficiencies of only 69.1% and 71.4% for particles in the 0.7–0.9 kDa and 1.0–1.5 kDa molecular weight ranges, respectively.

It should also be noted that the lowest molecular weight fraction (<0.15 kDa) was removed less efficiently in all systems, with removal efficiencies ranging from 2.4 to 40.4% for the conventional system, from 1.4 to 31.4% for the MF+NF system, and from 9.8 to 44.8% for the UF+NF system. The marked reduction in organic matter achieved with the integrated MF+NF and UF+NF systems significantly lowers the risk of disinfection by-product formation (Chaukura et al. 2020).

The removal of medium- and high-molecular-weight organic substances in the integrated membrane systems resulted in a marked increase in the proportion of

biodegradable dissolved organic carbon (BDOC) within the total organic carbon (TOC) fraction (Figure 12). In contrast, the conventional water treatment system showed only a negligible increase in the BDOC fraction. A high proportion of biodegradable organic matter in the treated water indicates that these compounds are readily available to microorganisms in the distribution system, thereby increasing the risk of secondary microbial growth within the water supply network (Ndiongue et al. 2005).

Costs analysis

The costs of water treatment in the conventional system included coagulation, filtration, and adsorption, with the unit costs for each stage summarized in Table 7.

The cost of coagulation in the conventional system depended primarily on the type of coagulant used and, to a lesser extent, on its dosage, with higher costs associated with pre-hydrolyzed coagulants.

In the integrated membrane systems (MF+NF and UF+NF), electricity consumption accounted for the largest share of total operational costs, exceeding 90%. The overall water treatment costs in these systems (Table 8) varied considerably, largely due to the need to heat the feed water to maintain stable permeate flux. During periods of declining feed water temperature, the

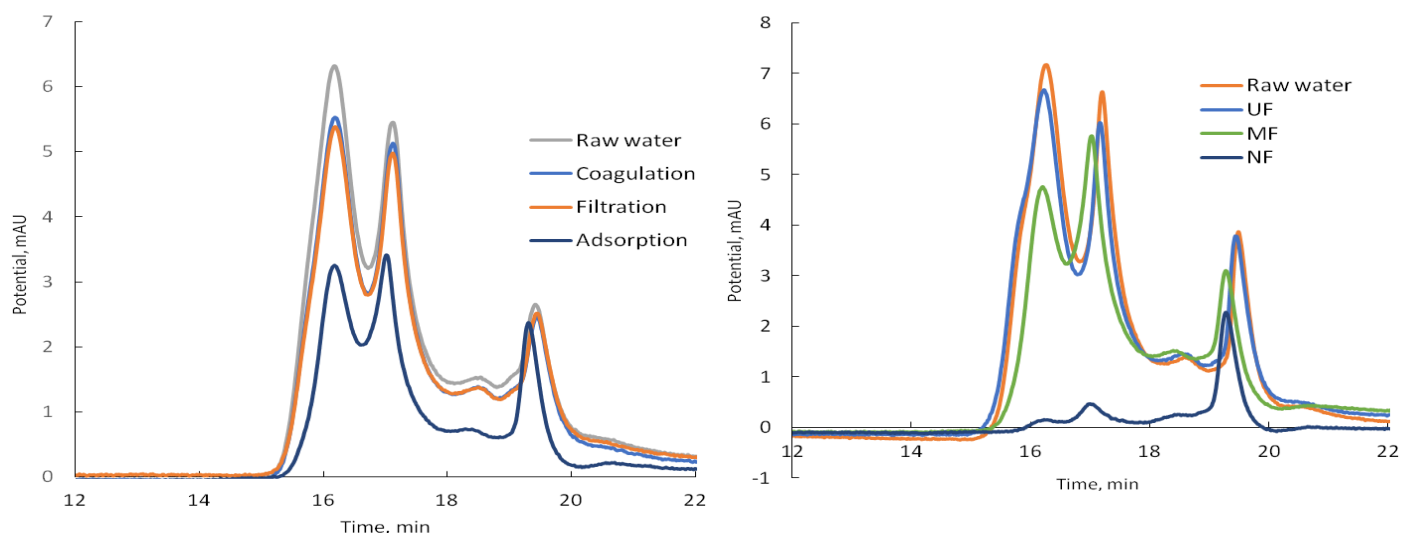


Fig. 10. Size exclusion chromatography of water treated by conventional processes (a) and membrane processes (b)

Table 7: Unit operational costs of the conventional water treatment system (euro/m³)

Process / operation	Water intake	Coagulation	Filtration	Adsorption
Unit cost	0.02	0.03–0.06	0.03	0.03

Table 8: Costs of integrated membrane processes (euro/m³)

Process	MF+NF	UF+NF
With heating (to 5°C)	7.70–15.20	4.73–9.23
Without heating	6.72	4.16

feed water was heated to 5 °C. This was intended to reduce the adverse effect of increasing water viscosity on membrane hydraulic performance.

The operational costs observed in this study were substantially higher than those reported by other researchers (Othman et al. 2021), likely due to the higher operating pressures applied during microfiltration and ultrafiltration. Additionally, the MF+NF system incurred higher operational costs than UF+NF system, reflecting the significantly lower permeate flux achieved with microfiltration.

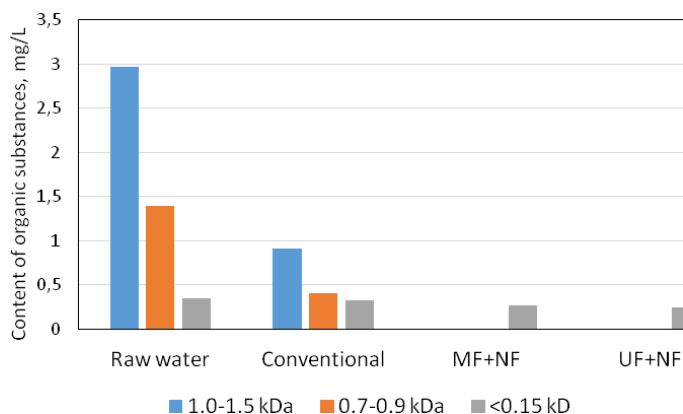
Eliminating the water heating step prior to MF and UF processes can considerably reduce treatment costs. However, even without water heating, the cost of the MF+NF system remains approximately 33 times higher than that of the conventional water treatment system, while the cost of the UF+NF system is approximately 20 times higher.

It should also be noted that these cost estimates are associated with considerable uncertainty due to the relatively small scale of the experimental setup.

Conclusions

The study demonstrated that:

- Integrated membrane systems (MF+NF and UF+NF) achieved significantly higher removal of organic matter, with average reductions exceeding 95% for color, 93% for DOC, and 93% for UV₂₅₄ absorbance, compared with 76%, 50%, and 61%, respectively, for the conventional treatment system.

**Fig. 11.** Average concentrations of organic substance fractions in raw water and in water treated by conventional and integrated membrane systems

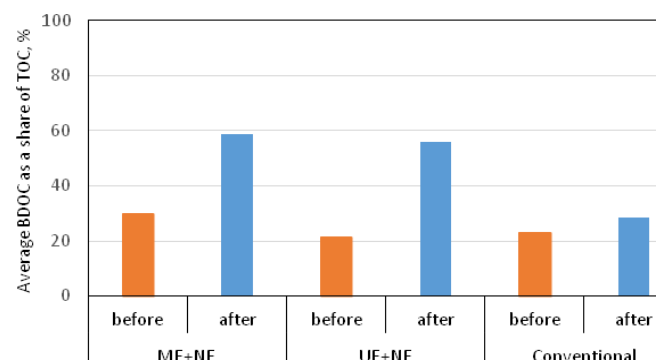
- Surface water treatment using integrated membrane systems removed biodegradable dissolved organic carbon (BDOC) more effectively than conventional treatment system.
- Integrated membrane systems removed microorganisms far more efficiently than the conventional treatment systems; water treated by the conventional system contained 15 and 5 times more cells than water treated by the UF+NF and MF+NF systems, respectively.
- Treatment using the integrated MF+NF and UF+NF systems reduced water salinity and achieved water softening, whereas the conventional treatment system increased water salinity.

Acknowledgments

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**Fig. 12.** Average proportion of BDOC in TOC for raw and treated water

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Porównanie skuteczności oczyszczania wód powierzchniowych z wykorzystaniem zintegrowanych ciśnieniowych procesów membranowych oraz systemów konwencjonalnych

Ciśnieniowe techniki membranowe są coraz częściej stosowane w uzdatnianiu wód naturalnych jako alternatywa dla konwencjonalnych procesów fizyczno-chemicznych. Sprawia to, że niezbędne staje się porównanie skuteczności obu tych podejść, szczególnie w warunkach długotrwałej eksploatacji. W fazie wstępnej badań przeprowadzono testy krótkookresowe, aby na podstawie właściwości separacyjnych i transportowych wyłonić najodpowiedniejsze membrany: do dalszych badań długofalowych wybrano dwie z trzech membran do mikrofiltracji (MF), dwie z trzech do ultrafiltracji (UF) oraz jedną z dwóch do nanofiltracji (NF). Podczas wspomnianych testów długookresowych oceniano efektywność oczyszczania wód powierzchniowych w zintegrowanych układach MF+NF oraz UF+NF, a następnie porównano ją ze skutecznością konwencjonalnego układu technologicznego obejmującego koagulację, sedymentację, filtrację oraz adsorpcję. Wyniki wskazują, że zintegrowane układy MF+NF lub UF+NF pozwalają na uzyskanie znacznie wyższej efektywności oczyszczania niż procesy konwencjonalne; przykładowo, redukcja rozpuszczonego węgla organicznego (DOC) wyniosła 93% w systemach membranowych wobec 50% w metodach konwencjonalnych. Procesy membranowe umożliwiają również wysoce skuteczne usuwanie specyficznych mikroorganizmów, co jest nieosiągalne przy zastosowaniu tradycyjnych metod uzdatniania. Ponadto zintegrowane układy MF+NF oraz UF+NF obniżają zasolenie wody i zapewniają jej zmiękczenie, podczas gdy systemy konwencjonalne wykazują tendencję do zwiększania zasolenia. Analiza kosztów eksploatacyjnych wykazała, że układ UF+NF jest o niemal 50% tańszy niż układ MF+NF. Niemniej jednak oba systemy oparte na technikach membranowych pozostają kilkukrotnie droższe od konwencjonalnego uzdatniania wody. Badania pokazały, że zastosowanie zintegrowanych procesów membranowych pozwala na uzyskanie wody o lepszej jakości niż w wypadku użycia procesów konwencjonalnych.